

TELCO



Data Analysis Final Project

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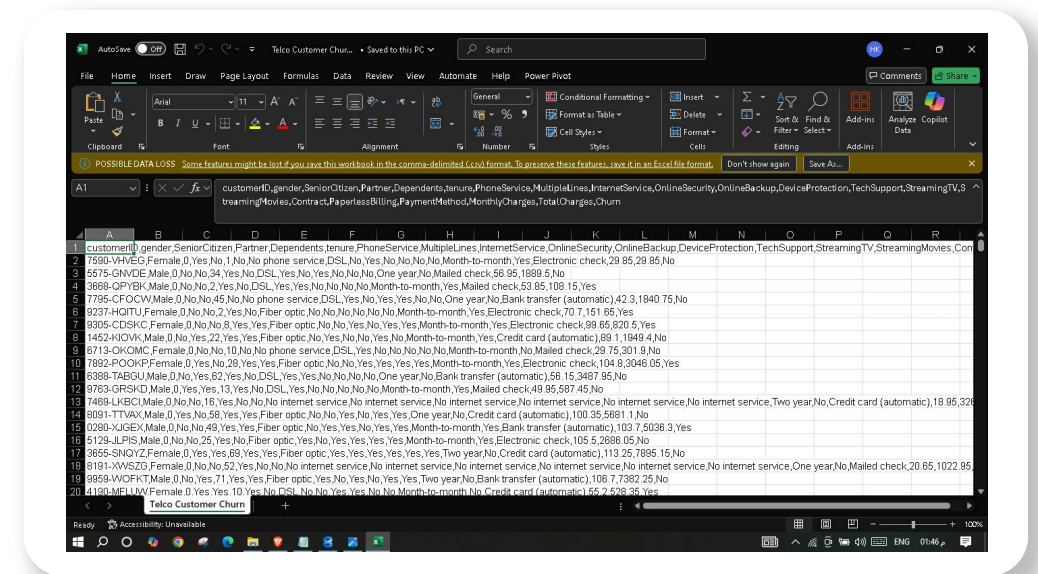
Telco Customer Churn Analysis

This project provides an in-depth analysis of customer churn in a fictional telecommunications service provider company. The analysis focuses on understanding the demographics and account information of high-risk customers to identify factors contributing to churn. This project is intended to help businesses make data-driven decisions to improve customer retention.

Project Overview:

The goal of this analysis is to:

- Analyze customer churn data to understand key factors associated with churn.
- Visualize churn rates to make it easy for the decision makers to take an action.
- Provide insights and recommendations to reduce churn.

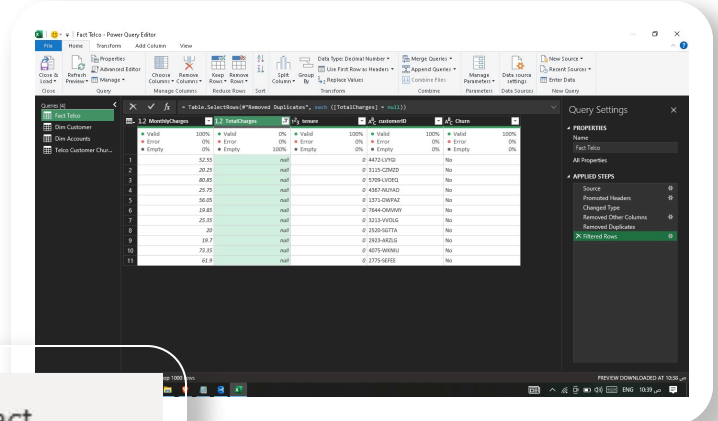
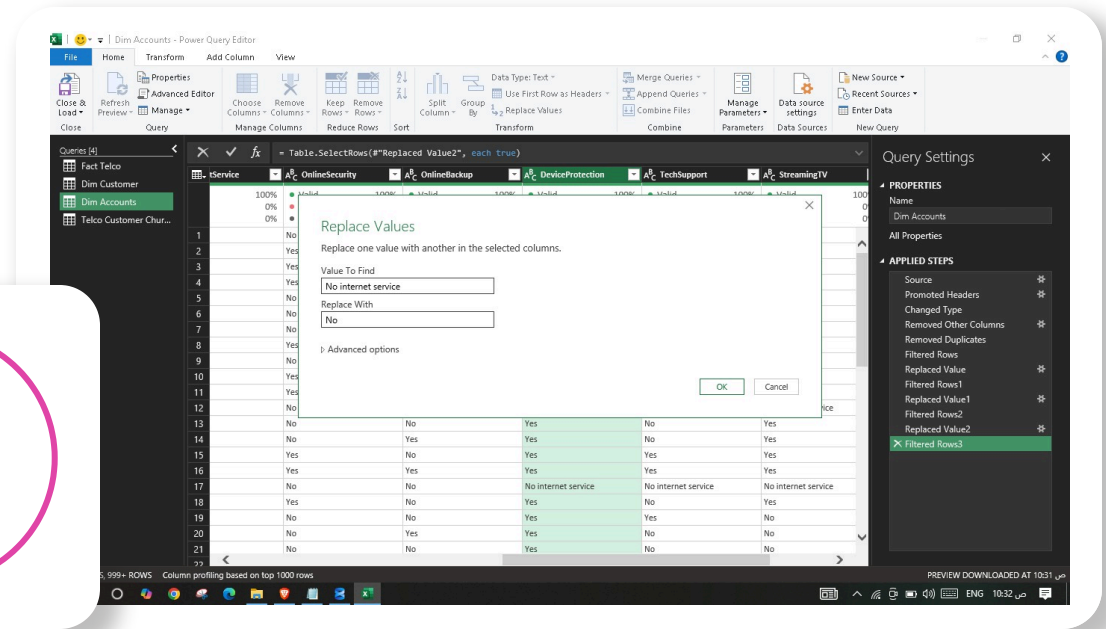
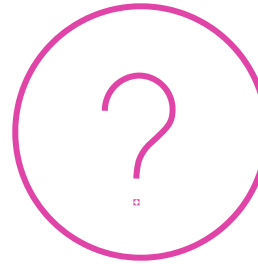


EDA Questions:

- Q1: How many customers have subscribed to date?
- Q2: How many have unsubscribed from our services?
- Q3: What is the number of current customers?
- Q4: What is the churn rate?
- Q5: What is the average Tenure In Months?
- Q6: What is the annual revenue?
- Q7: What is the number of subscribed and unsubscribed customers per their demographic information?
- Q8: What are services the subscribed and unsubscribed customers were using?
- Q9: Which factors have the strongest correlation with churn?

Data Cleaning and Transformation:

- Total Charges column has null values that need to be handled, it has 11 nulls, and we replaced them by zero and this is done because we find tenure values equal to zero.
- Replace [0] by [No] and replace [1] by [Yes] in Senior Citizen.
- Replace [No internet service] by [No].



- > Dim Contract
- > Dim Customer
- > Dim Services
- > Fact

Converting into star schema



```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column             Non-Null Count  Dtype
---  -
 0   customerID         7043 non-null   object
 1   gender             7043 non-null   object
 2   SeniorCitizen      7043 non-null   int64
 3   Partner            7043 non-null   object
 4   Dependents         7043 non-null   object
 5   tenure             7043 non-null   int64
 6   PhoneService       7043 non-null   object
 7   MultipleLines       7043 non-null   object
 8   InternetService    7043 non-null   object
 9   OnlineSecurity     7043 non-null   object
10  OnlineBackup       7043 non-null   object
11  DeviceProtection   7043 non-null   object
12  TechSupport        7043 non-null   object
13  StreamingTV        7043 non-null   object
14  StreamingMovies    7043 non-null   object
15  Contract           7043 non-null   object
16  PaperlessBilling   7043 non-null   object
17  PaymentMethod      7043 non-null   object
18  MonthlyCharges     7043 non-null   float64
19  TotalCharges       7043 non-null   object
20  Churn              7043 non-null   object
dtypes: float64(1), int64(2), object(18)
```

```
print(df.isnull().sum())

customerID    0
gender         0
SeniorCitizen  0
Partner        0
Dependents     0
tenure         0
PhoneService   0
MultipleLines   0
InternetService 0
OnlineSecurity  0
OnlineBackup    0
DeviceProtection 0
TechSupport     0
StreamingTV     0
StreamingMovies 0
Contract        0
PaperlessBilling 0
PaymentMethod   0
MonthlyCharges  0
TotalCharges    0
Churn           0
```

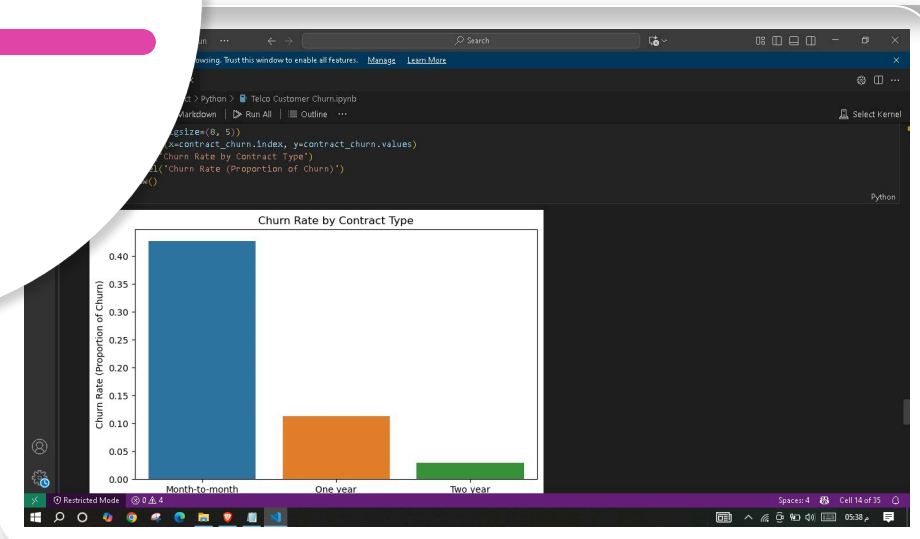
Python

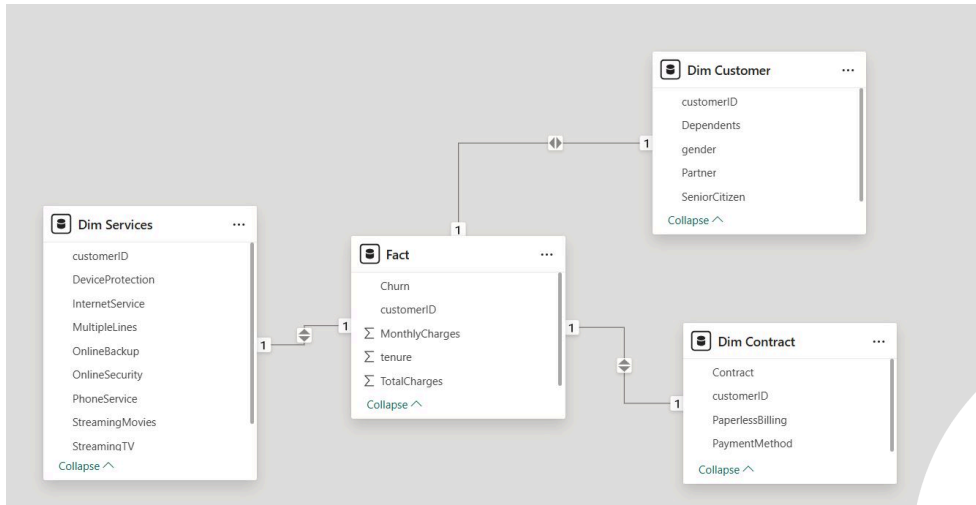
```
##Some Statcal For Some Columns .

print("Descriptive Statistics for Numerical Columns")
print(df[['tenure', 'MonthlyCharges', 'TotalCharges']].describe())

--- Descriptive Statistics for Numerical Columns ---
   count    7032.000000    7032.000000    7032.000000
   mean      32.421786      64.798280     2283.308441
   std       24.545260      38.085974     2266.771362
   min        1.000000      19.250000      18.000000
   25%        9.000000     35.507500     481.450000
   50%       29.000000     70.350000    1397.475000
   75%       55.000000     89.862500    3794.737500
   max       72.000000    118.750000    8084.000000

##Histogram TO Show The Distribution On Numerical Features .
```





```
-- Number Of Columns in each Table
SELECT COUNT(*) AS Total_Fact_Records FROM [Fact Telco]
SELECT COUNT(*) AS Total_Customer_Records FROM [Dim Accounts]
SELECT COUNT(*) AS Total_Accounts_Records FROM [Dim Customer]
```

Diagram and SQL Queries

```
-- Describe Of Customer Payment
SELECT
    AVG(MonthlyCharges) AS Avg_Monthly_Charges,
    MAX(MonthlyCharges) AS Max_Monthly_Charges,
    MIN(MonthlyCharges) AS Min_Monthly_Charges,
    AVG(Total_Charges) AS Avg_Total_Charges,
    AVG(tenure) AS Avg_Tenure_Months
FROM [Fact Telco]
GO
```

```
33
34 --Number Of Churned Customer By PaymentMethod
35 SELECT
36     DC.PaymentMethod,
37     SUM(CASE WHEN FT.Churn = 'True' THEN 1 ELSE 0 END) AS Churned_Count,
38     COUNT(FT.customerID) AS Total_Count,
39     (CAST(SUM(CASE WHEN FT.Churn = 'True' THEN 1 ELSE 0 END) AS FLOAT) / COUNT(FT.customerID)) * 100 AS Churn_Rate
40 FROM [Fact Telco] FT
41 JOIN [Dim Customer] DC
42 ON FT.customerID = DC.customerID
43 GROUP BY DC.PaymentMethod
44 ORDER BY Churn_Rate DESC
45 GO
46
```

```
-- Number Of Churned Customers From Total Customers
SELECT
    SUM(CASE WHEN Churn = 'True' THEN 1 ELSE 0 END) AS Total_Churned,
    COUNT(*) AS Total_Customers,
    (CAST(SUM(CASE WHEN Churn = 'True' THEN 1 ELSE 0 END) AS FLOAT) / COUNT(*)) * 100 AS Churn_Rate_Percentage
FROM [Fact Telco]
GO
```

```
-- Number Of Churned Customer By InternetService
SELECT
    DA.InternetService,
    SUM(CASE WHEN FT.Churn = 'True' THEN 1 ELSE 0 END) AS Churned_Count,
    COUNT(FT.customerID) AS Total_Count,
    (CAST(SUM(CASE WHEN FT.Churn = 'True' THEN 1 ELSE 0 END) AS FLOAT) / COUNT(FT.customerID)) * 100 AS Churn_Rate
FROM [Fact Telco] FT
JOIN [Dim Accounts] DA
ON FT.customerID = DA.customerID
GROUP BY DA.InternetService
ORDER BY Churn_Rate DESC
GO
```



gender

All

Total No. Customers

7043

Active Customers

5174

Churn Customers

1869

Churn Rate

27%

Average Tenure In Months

32

Total Revenue

16M

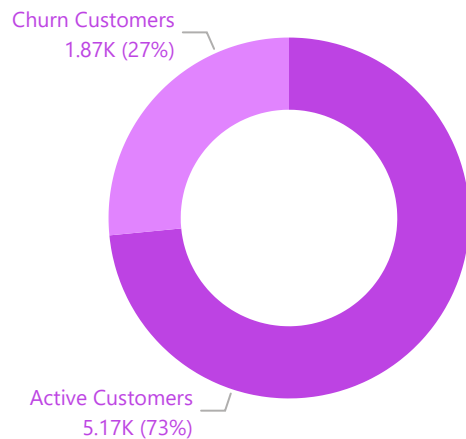
Overview

customer

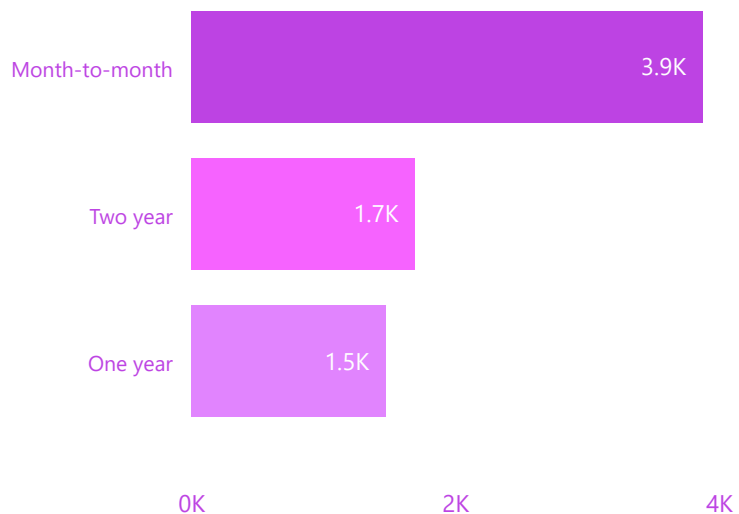
services

contract

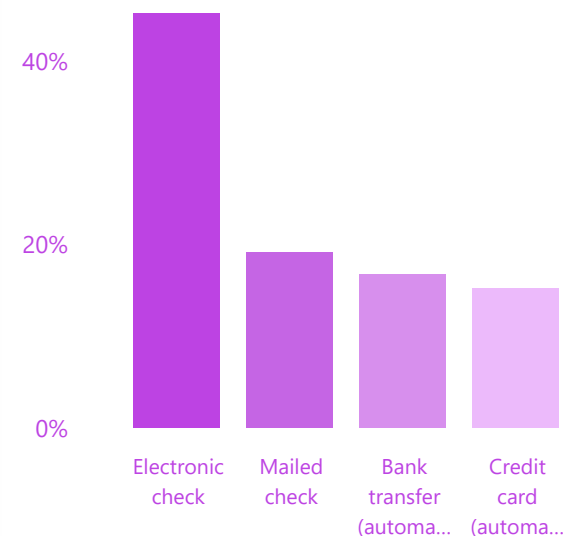
Active Customers VS Churn Customers



No. of Customers by Contract



Churn Rate by PaymentMethod



TELCO

Total No. Customers

7043

Active Customers

5174

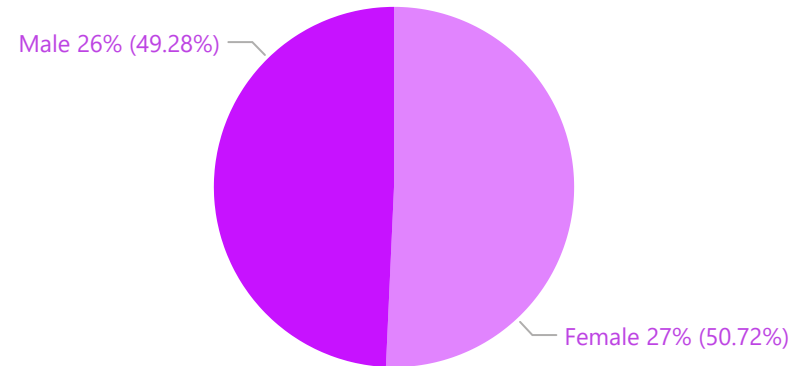
Churn Customers

1869

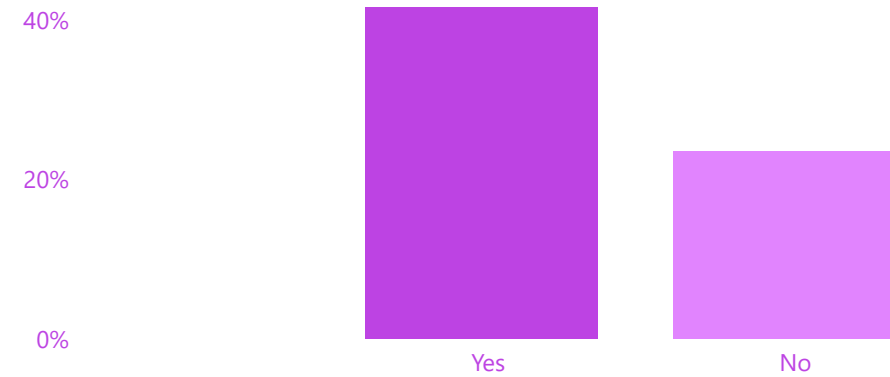
Churn Rate

27%

Churn Rate by gender



Churn Rate by SeniorCitizen



Churn Rate by Partner and Dependents



Total No. Customers

7043

Active Customers

5174

Churn Customers

1869

Senior Customers

1142

Seniors Churn Rate

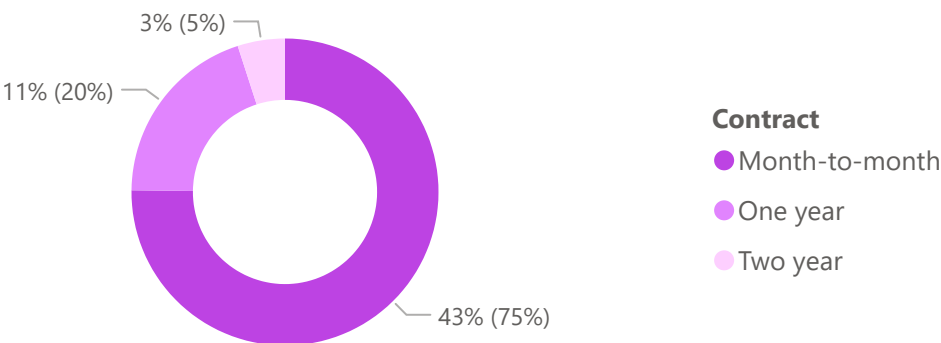
25%

Senior Citizen

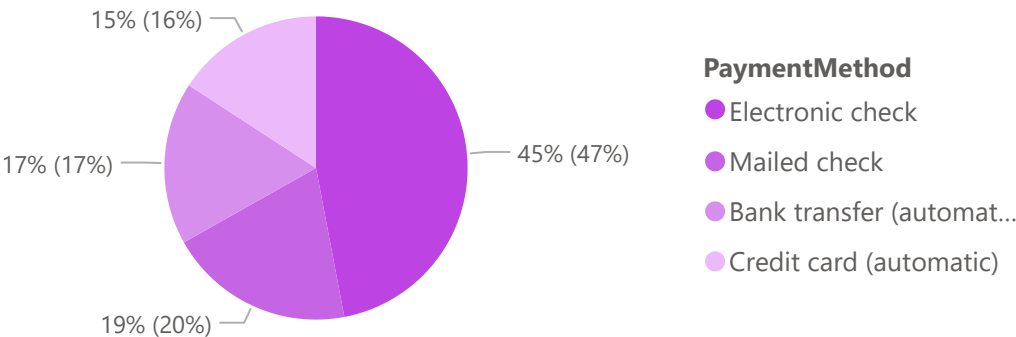
No

Yes

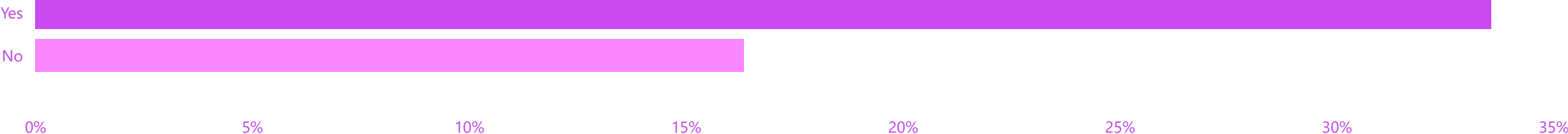
Churn Rate by Contract

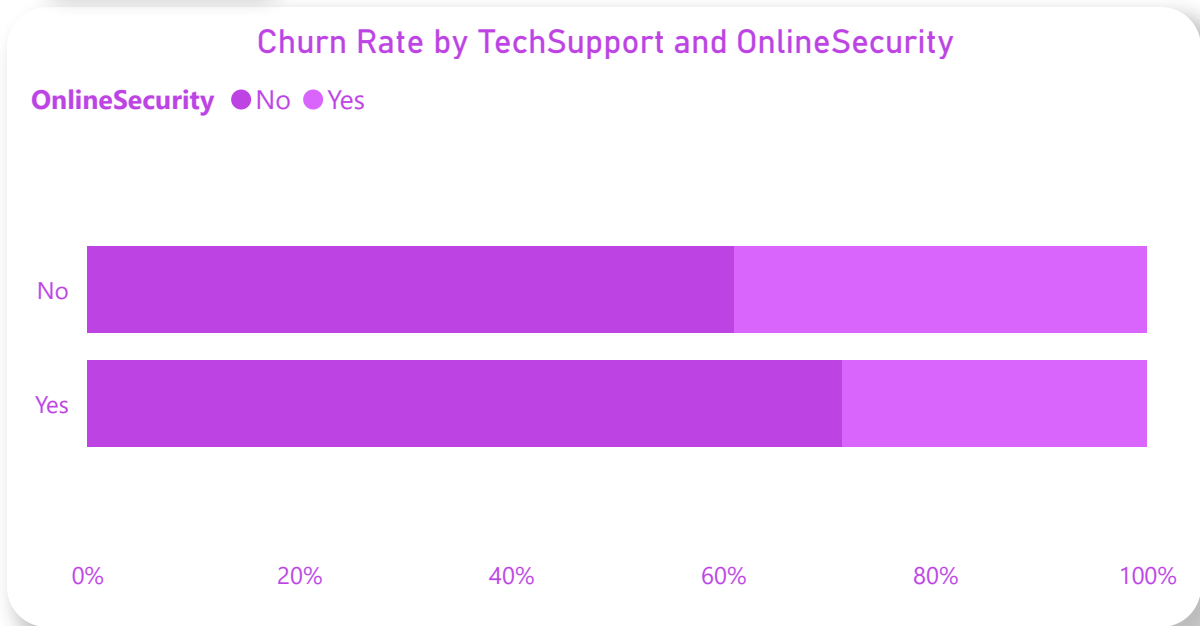
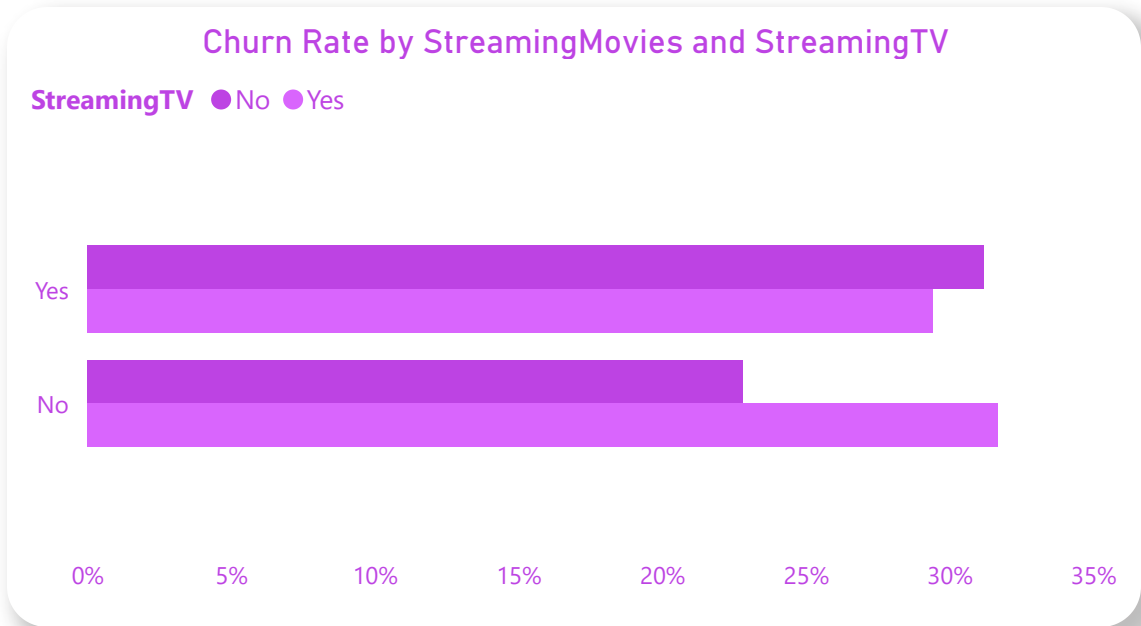
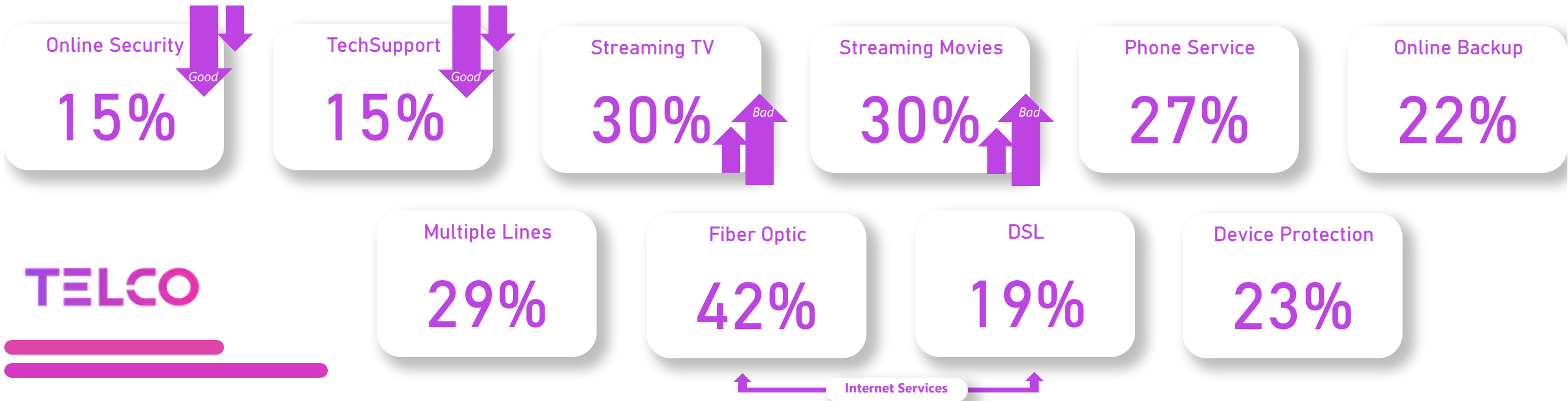


Churn Rate by PaymentMethod



Churn Rate by PaperlessBilling





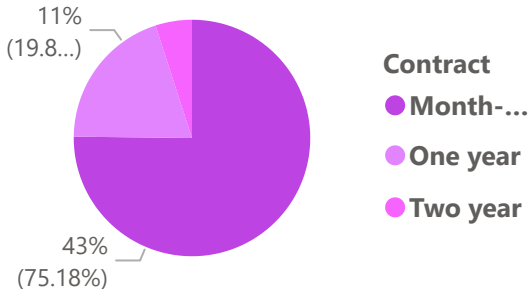
Observations:

1. The churn rate is relatively high, especially among customers on month-to-month contracts.
2. Customers with short tenure (less than one year) are more likely to leave the service.
3. Customers with higher monthly charges (particularly those using Fiber Optic + add-on services) show a higher churn tendency.
4. Month-to-month contracts have the highest churn, while one-year and two-year contracts retain customers better.
5. Customers paying via Electronic Check have a significantly higher churn rate compared to other payment methods.
6. Senior citizens demonstrate a higher churn rate than younger customer groups.
7. Customers subscribed to additional services (e.g., Online Security, Tech Support) have a lower churn rate.

Recommendations:

1. Encourage new customers to subscribe to yearly or two-year contracts instead of monthly ones by offering discounts or special promotions.
2. Reconsider pricing for customers with high monthly charges, especially Fiber Optic users, to improve perceived value.
3. Promote adoption of services that reduce churn (e.g., Tech Support, Online Security) through bundled offers or free trials.
4. Provide additional support and simplified communication tailored to senior citizens.
5. Incentivize customers to switch from Electronic Check to more secure and automatic payment methods (Credit Card, Bank Transfer).
6. Launch a loyalty program with extra benefits for customers with short tenure (< 6 months) to improve early retention.

Churn Rate by Contract



Churn Rate by PaymentMethod

